UNIVERSITY OF MAURITIUS
SIS 3204Y(5) – NATURAL LANGUAGE PROCESSING

MOCK PAPER 3 – Weeks 1 to 8 (Question Paper with Model Answers)
Total: 50 marks | Suggested duration: 60 minutes

Instructions: Answer ALL questions. Marks are shown beside each question. The week each question comes from is shown in the question header. Model answers are given under each question in bullet points; the mark for a point is shown in brackets, e.g. (1). Cover the answers while practising. Equivalent correct answers should receive credit.

Question 1 | Week 1 (Revision Q9) [5 marks]

What is a Transformer and why is it important in modern NLP?

Answer

- **Definition:** a neural network architecture designed to process and model sequential language data. (1)
- **Attention:** uses self-attention to work out the relationships between all words in a sentence at the same time. (1)
- **Long-range dependencies:** captures global context much better than RNNs/LSTMs. (1)
- **Parallel processing:** operations on tokens can run in parallel, which improves training efficiency. (1)
- **Foundation of modern AI:** the core architecture behind BERT, GPT and modern LLMs. (1)

Question 2 | Week 2 (Revision Q16) [4 marks]

Using TWO examples, explain how removing stop words can change the meaning of a sentence.

Answer

- **Example 1:** “Where is the nearest bank?” → “nearest bank”. The question word “where” is lost, so it is no longer a question about location. (2)
- **Example 2:** “She is not happy.” → “happy”. The negation “not” is removed, so the meaning becomes the opposite. (2)

Question 3 | Week 3 (Revision Q12) [6 marks]

What is Word2Vec? State the distributional hypothesis and explain how Word2Vec learns word representations.

Answer

- Word2Vec is a family of **unsupervised** models that learn dense, low-dimensional word vectors (embeddings) from large text by using the context in which words appear (Mikolov et al., Google, 2013). (1)
- **Distributional hypothesis:** words that occur in similar contexts tend to have similar meanings. (1)
- Learning: take a large corpus, preprocess it (tokenise, lowercase), then slide a **context window** over sentences to make (target, context) pairs. (1)
- Objective: predict context words from a target word (**Skip-Gram**) or the target from context words (**CBOW**); optimised with stochastic gradient descent. (1)
- Result: each word gets a dense vector (typically 50–300 dimensions); similar words are close together; no manual labelling needed. (1)
- It captures relationships such as king − man + woman ≈ queen and Paris − France + Italy ≈ Rome. (1)

Question 4 | Week 4 (Part B Q6 and Q4) [6 marks]

(a) Excellent training performance but poor performance on new text: identify the problem and give three remedies. (4 marks)

(b) Probabilities: Positive 0.72, Neutral 0.18, Negative 0.10. What is predicted and why? (2 marks)

Answer

(a)

- Problem: **overfitting** – the model learns the training data (and noise) too closely and does not generalise. (1)
- Any three remedies: dropout, more data, simpler model, regularisation, early stopping, data augmentation. (3)

(b)

- The predicted class is **Positive**. (1)
- Because 0.72 is the **highest** output probability. (1)

Question 5 | Week 5 (Sample Q10) [10 marks]

RNN, LSTM and GRU have been widely used for NLP. Explain the limitations of recurrent architectures that motivated the move towards attention and Transformers.

Answer

- **Sequential processing:** RNN, LSTM and GRU read text one step at a time and each step depends on the previous one, so little parallel processing is possible. (2)
- **Long-term dependencies:** a vanilla RNN has vanishing gradients, so it struggles to learn links between distant words. (2)
- LSTM and GRU **improve** this with gates, but they are still recurrent and can be inefficient for very long texts. (1)
- **Attention:** lets the model decide which words/positions are most relevant to each other, so distant information is reached more directly. (2)
- **Transformers** use self-attention and remove token-by-token recurrence. (1)
- This gives greater **parallelisation and scalability** for large datasets and long sequences. (2)

Question 6 | Week 6 (Set 1 Q4) [8 marks]

Define self-attention. How is it different from encoder-decoder attention?

Answer

- **Self-attention:** each token looks at other tokens in the **same sequence** to build its contextual representation. (2)
- In self-attention, the Query, Key and Value all come from the **same sequence**. (2)
- **Encoder-decoder attention:** a decoder query attends to representations from the **encoder** (a different sequence). (2)
- So Q, K and V are derived from the same sequence in self-attention. (1)
- Result: each token representation includes information from other relevant tokens, giving a contextual representation. (1)

Question 7 | Week 7 (Set 1 Q5) [6 marks]

A software company is developing three NLP applications:

Application A: Identify names of people, organisations and locations in customer emails.

Application B: Generate a response to a customer's enquiry.

Application C: Translate customer emails from English into French.

For each application: a) Recommend an appropriate architecture (encoder-only, decoder-only or encoder–decoder). (3 marks) b) Justify the choice based on language processing/generation. (3 marks)

Answer

Application A – Named entity recognition

- **Encoder-only** (e.g. BERT). (1)
- Bidirectional contextual representations suit entity tagging. (1)

Application B – Response generation

- **Decoder-only** (e.g. GPT). (1)
- Autoregressive generation produces the response token by token. (1)

Application C – Translation

- **Encoder–decoder** (e.g. T5 or Marian). (1)
- The encoder represents the source text; the decoder generates the translation using causal self-attention and cross-attention. (1)

Question 8 | Week 8 (Tutorial Set A Q4) [5 marks]

(a) Differentiate between zero-shot and few-shot prompting. (2 marks)

(b) Construct a few-shot prompt for classifying customer messages as Order, Complaint or Enquiry. Include two demonstrations and one new message to classify. (3 marks)

Answer

(a)

- **Zero-shot:** gives instructions only, without task examples. (1)
- **Few-shot:** includes labelled input–output examples in the prompt. (1)

(b) Example prompt

- "Classify each message as Order, Complaint or Enquiry." (1 – instruction and labels)
- Message: I would like to purchase two dresses. Label: Order / Message: My parcel arrived damaged. Label: Complaint (1 – two correct demonstrations)
- Message: Do you have this item in size XL? Label: → expected output **Enquiry**. (1 – new input and correct label)

Marks Summary

Question	Week	Source	Marks
1	1	Revision Q9	5
2	2	Revision Q16	4
3	3	Revision Q12	6
4	4	Part B Q6 and Q4	6
5	5	Sample Q10	10
6	6	Set 1 Q4	8
7	7	Set 1 Q5	6
8	8	Tutorial Set A Q4	5
		TOTAL	**50**